Model training conversion

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References

After completing the tutorial content of dataset annotation, we can use the motherboard to start training the model.

```
This tutorial only introduces the model training and conversion of CLI. You can
refer to the official website to modify the Python case
```

1. Model training

Use CLI command to train the model directly: copy the yolo11n.pt file to the directory where the configuration file is located, and then open the terminal in the directory where the configuration file is located:

```
cd /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data
```

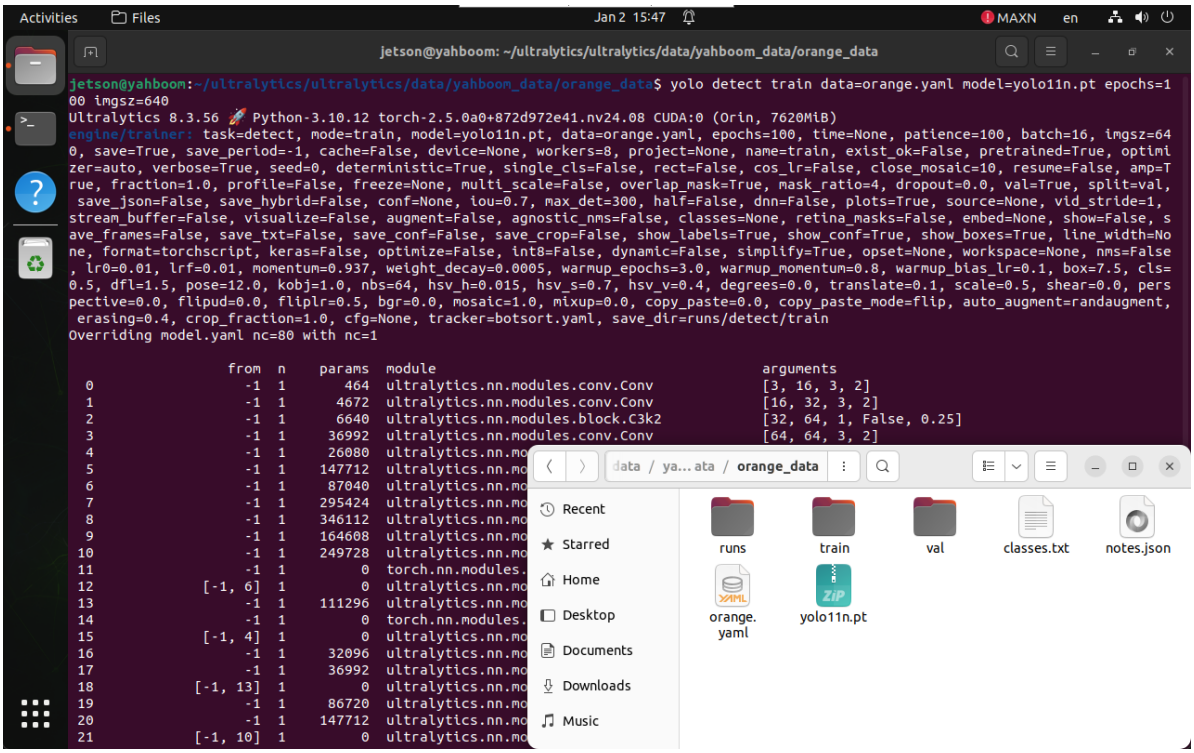
```
yolo detect train data=orange.yaml model=yolo11n.pt epochs=100 imgsz=640
```

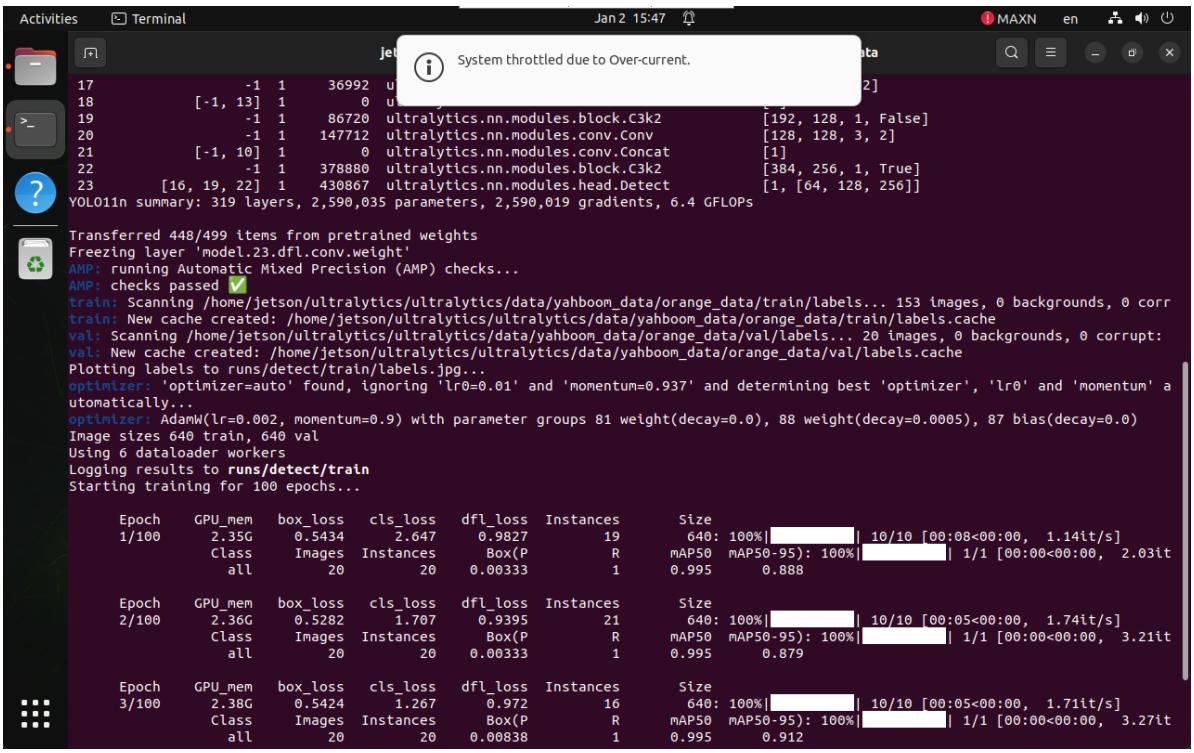
`data`: Dataset configuration file

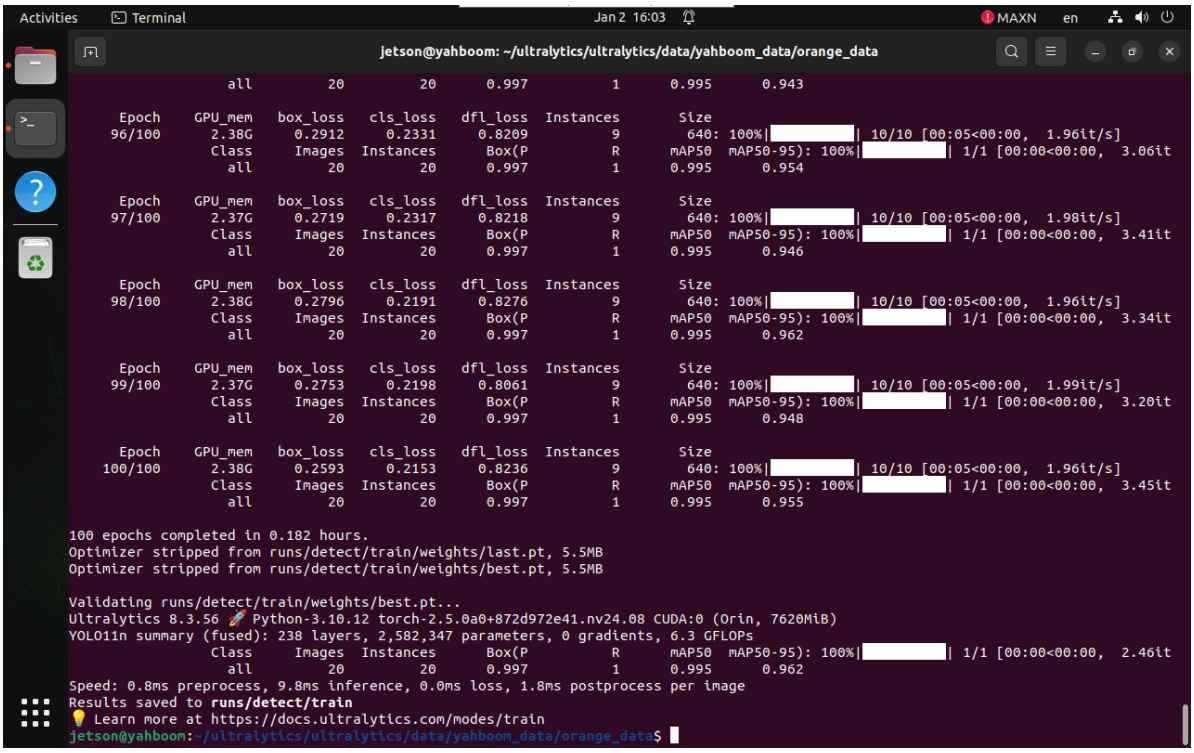
`model`: Pre-trained model file

`epochs`: Number of training rounds

`imgsz`: Input specified image size







2. Model conversion

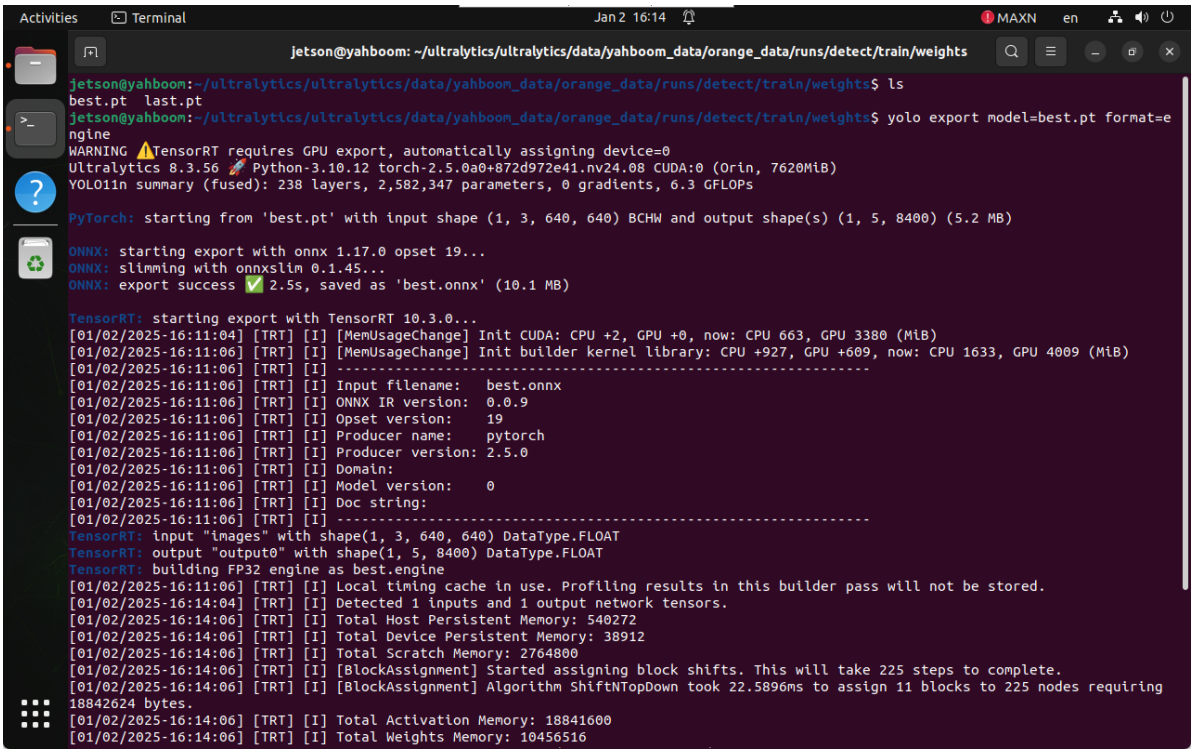
The final model will be generated in the runs folder: generally select the best.pt file for use

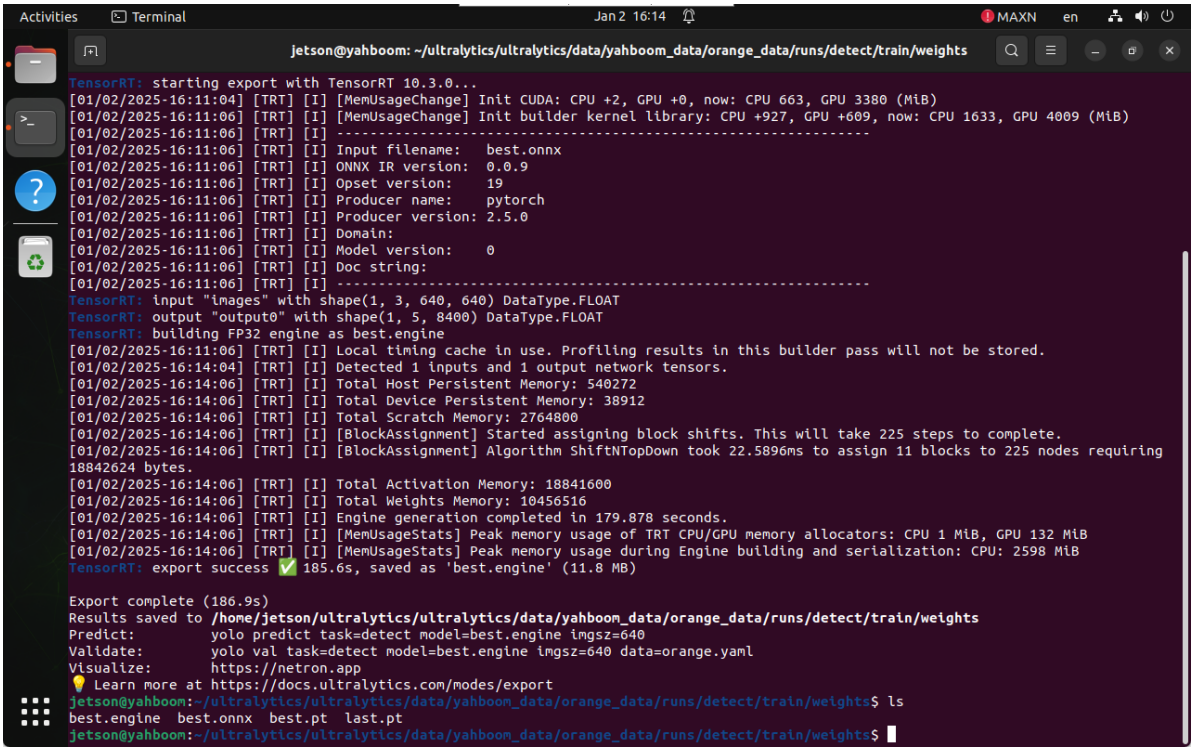
```
/home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights
```

Convert the PyTorch format model to TensorRT:

```
cd /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights
```

```
yolo export model=best.pt format=engine
```





References

https://docs.ultralytics.com/modes/train/